Last updated: May 9, 2026

Hyeonbeen Lee

leehyeonbeen@vt.edu • Homepage • Scholar • LinkedIn • GitHub

EDUCATION

Virginia Tech, Blacksburg, VA, USA — Aug. 2026 — May 2031 (Expected)
Ph.D. in Mechanical Engineering (Advisor: Simon Stepputtis)
Research Focus: Robot Learning, Neurosymbolic AI, Physics-aware ML

Kyung Hee University, Seoul, South Korea — Mar. 2022 — Feb. 2024
M.Eng. in Mechanical Engineering (Advisor: Jin-Gyun Kim) — GPA: 3.93/4.0
Thesis: *'Composite neural network with differential propagation for modeling impulsive nonlinear dynamic systems'*

Kyung Hee University, Seoul, South Korea — Mar. 2015 — Feb. 2022
B.Eng. in Mechanical Engineering (Advisor: Jin-Gyun Kim, Shinkyu Jeong) — (Upper Div.: 3.88/4.0) GPA: 3.52/4.0
Thesis: *'Data-driven aerodynamic coefficient prediction using deep neural network and PARSEC airfoil parameterization'*
(∗ Military leave of absence: May 2017 — Feb. 2019)

PUBLICATIONS

Journal Papers

4. **H. Lee**, M. Jung, T.K. Yeu, J.B. Han, D. Park, J.G. Kim (2026), "Frequency-aware decomposition learning for sensorless wrench forecasting on a vibration-rich hydraulic manipulator", submitted to *IEEE/ASME Transactions on Mechatronics.* [arXiv] [code]
3. B. Koo, M. Jung, **H. Lee**, T. Yeu, J.G. Kim, J.B. Han, Y. Lee, D. Park (2026), "Buoyancy-integrated hybrid reaction force estimation method with real-time haptic feedback for underwater hydraulic manipulation", submitted to *IEEE Robotics and Automation Letters.*
2. **H. Lee**, S. Han, H.S. Choi, J.G. Kim (2024), "cNN-DP: Composite neural network with differential propagation for impulsive nonlinear dynamics", *Journal of Computational Physics (WoS IF Top 2.5% in Physics, Mathematical)*, 496, 112578. [link] [code]
1. S. Han*, G.E. Jeong*, **H. Lee**, W.S. Choi, J.G. Kim (2023), "Multi-body dynamics model for spent nuclear fuel transportation system under normal transport test conditions", *Nuclear Engineering and Technology (WoS IF Top 13.7% in Nuclear Science & Technology)*, 55(11), 4125-4133. (*Co-first authors) [link]

Conferences & Proceedings

6. J.H. Han, J.B. Han, S.S. Kim, M.H. Kim, Y.H. Kim, **H. Lee**, J.G. Kim, T.K. Yeu. "Digital twin model of underwater construction robot for real-time grinding simulation", *7th International Conference on Multibody System Dynamics*, Madison, WI, USA, Jun. 2024 [link]
5. **H. Lee**, J. Han, T.K. Yeu, J.G. Kim. "Real-time multi-horizon reaction force forecasting of marine robot using interpretable Transformer", *Korean Society of Mechanical Engineers* (Oral Presentation), Incheon, South Korea, Nov. 2023 [link]
4. **H. Lee,** S. Han, H.S. Choi, J.G. Kim. "Meta-modeling of nonlinear impulsive dynamics using composite neural network model with differential propagation", *Korean Society of Mechanical Engineers* (**Excellent Paper Award**, Oral Presentation, ∗ Extended works from IMAC 2023), Busan, South Korea, May 2023. [link]
3. **H. Lee,** S. Han, H.S. Choi, J.G. Kim. "Composite neural network with differentiation propagation for modeling nonlinear impulsive dynamics", *Conference on Dynamics and Control, Korean Society of Mechanical Engineers* (Oral Presentation), Jeju, South Korea, Mar 2023 [link]
2. **H. Lee,** S. Han, H.S. Choi, J.G. Kim. "Composite neural network framework for modeling impulsive nonlinear dynamic responses", *41st International Modal Analysis Conference (IMAC)* (Oral Presentation), Austin, TX, USA, Feb. 2023 [link]
1. **H. Lee**, J.G. Kim. "Development of multibody dynamics trailer model using normal transportation test data and DNN based surrogate model generation", *Korean Society for Noise and Vibration Engineering* (Oral Presentation), Jeju, South Korea, Dec 2022

AWARDS & HONORS

- **Excellent Paper Award** — Korean Society of Mechanical Engineers, 2023
  - Awarded for a first-author presentation on physics-aware modeling of nonlinear dynamics.
- **Brain Korea 21 Plus Scholarship** — Graduate School, Kyung Hee University, 2022 — 2023
  - Received $16K in graduate funding during M.Eng. studies.
- **Graduate Assistantship** — Graduate School, Kyung Hee University, 2022 — 2023

- Received $11K through a selective graduate assistantship during M.Eng. studies.

- **Academic Superiority Scholarship** Dept. of Mechanical Engineering, Kyung Hee University, Spring 2021
  - Ranked 1st in the GPA-based departmental merit review and received a full tuition waiver valued at $3K.

RESEARCH EXPERIENCES

Graduate School, Kyung Hee University Postgraduate Researcher
Modeling & Simulation Lab., advised by *Jin-Gyun Kim* Mar. 2024 — Apr. 2026

- Combined decomposition-based probabilistic modeling, frequency-aware architectures, and proprioception-to-wrench transfer learning to sensorlessly forecast contact- and vibration-rich wrench of a robotic manipulator. Submitted a first-authored manuscript to the *IEEE/ASME Transactions on Mechatronics (T-MECH)*.
- Accelerated and improved the force model for real-time deployment in the control loop for KRISO. Independently secured the project for a year.

Kim Jaechul Graduate School of AI, KAIST Research Engagement
Cognitive Learning for Vision & Robotics Lab., advised by *Joseph J. Lim* Sep. 2023 — Nov. 2023

- Conducted a literature review on skill prior regularized reinforcement learning (SPiRL) and delivered a 90-minute technical presentation to lab members and faculty.

Graduate School, Kyung Hee University Master's Student
Modeling & Simulation Lab., advised by *Jin-Gyun Kim* Mar. 2022 — Feb. 2024

- Proposed a composite neural network architecture for learning complex dynamics from multi-order derivatives. Published a first-author article in the *Journal of Computational Physics (JCOMP)*.
- Designed a probabilistic frequency-aware model to forecast contact-rich wrench of a hydraulic manipulator for the Korea Research Institute of Ships and Ocean Engineering (KRISO).
- Developed a neural metamodel of real-world vehicle dynamics with high-fidelity physics simulation for the Korea Atomic Energy Research Institute (KAERI). Published a co-authored article in the *Nuclear Engineering and Technology (NET)*.

Department of Mechanical Engineering, Kyung Hee University Undergraduate Researcher
Modeling & Simulation Lab., advised by *Jin-Gyun Kim* Jan. 2021 — Feb. 2022

- Investigated data-driven methods for flexible multibody dynamics as part of a government-funded research project on vehicle dynamics simulation.

SKILLS

- **Programming Languages:** Python · MATLAB · C++ · LaTeX
- **Software & Frameworks:** PyTorch · Git · Docker · ROS · RecurDyn · Gazebo · MuJoCo · OpenAI API
- **Scientific Computing:** Flexible multibody dynamics · Finite element methods · Numerical analysis
- **Natural Languages:** Korean (Native) · English (TOEFL iBT 111/120) · Japanese (Advanced proficiency)

INVITED SEMINARS

2. "*On the data-driven modeling of contact- and vibration-rich dynamics*", invited seminar, IMBD Lab, Chungnam National University, Daejeon, South Korea, Apr. 16, 2026 [keynote]
1. "*On the data-driven modeling of contact- and vibration-rich dynamics*", invited seminar, Modeling & Simulation Lab, Kyung Hee University, Yongin, South Korea, Apr. 14, 2026 [keynote]

SELECTED PROJECTS

- Segment-Anybody, at *Dept. of AI, Kyung Hee University* [code]
  - Fine-tuned Meta's Segment-Anything Model (SAM) for detailed human body parts segmentation in human-robot interaction.
- Deep seq2seq from scratch, *Personal project.* [code]
  - Scratch-implemented deep sequence-to-sequence models, from RNN to Transformer variants.
- ROS SLAM on Apple Silicon devices, *Personal project.* [code]
  - Addressed ROS TurtleBot3 SLAM and Gazebo compatibility issues on the Apple Silicon MacOS devices.
- RecurDyn ProcessNet with Python, at *Modeling & Simulation Lab.* [code]
  - Automated and parallelized flexible multibody dynamics (FMBD) simulations of RecurDyn.

TEACHING & MENTORSHIP

Research Mentor Modeling & Simulation Lab., Kyung Hee University
— Guiding junior graduate students on robotics and machine learning. Mar. 2025 — Present

Teaching Assistant, ME380 Systems Dynamics Dept. of Mechanical Engineering, Kyung Hee University
— Lectured on multibody dynamics with theoretical derivation and its computational implementation. Mar. 2022 — Jun. 2023

COMMUNITY SERVICE

Lead Administrative Assistant Dept. of Mechanical Engineering, Kyung Hee University, 2022 — 2024
48th College of Engineering Student Council Kyung Hee University, Feb 2019 — Jan. 2020
ROK-U.S. Combined Marine Corps Interpreter 1st Marine Div., ROKMC, Sep 2017 — Feb. 2019

REFERENCES

- **Jin-Gyun Kim**, Ph.D.

  Associate Professor at Department of Mechanical Engineering, Kyung Hee University (jingyun.kim@khu.ac.kr)
- **Hee-Sun Choi**, Ph.D.

  Assistant Professor at Department of Artificial Intelligence and Robotics, Sejong University (heesunchoi@sejong.ac.kr)
- **Seongji Han**, Ph.D.

  Assistant Professor at Department of Mechatronics Engineering, Chungnam National University (seongji.han@cnu.ac.kr)